Portfolio Risk Analysis

Artykzhan Ospan

Introduction

This report analyzes a growth oriented ETF portfolio with 70% stocks and 30% bonds.

The goal is to evaluate return, risk, diversification, and market stress performance from 2010 to 2024.

Portfolio Design

The baseline portfolio uses 70% stock exposure and 30% bond exposure.

Stocks provide growth exposure, while bonds add diversification.

ETF	Weight	Role
SPY	50%	Broad U.S. equity
QQQ	15%	Growth equity
IWM	5%	Small cap equity
IEF	20%	Intermediate Treasury bonds
TLT	10%	Long term Treasury bonds

Data and Methodology

The analysis uses daily adjusted price data for SPY, QQQ, IWM, IEF, and TLT from 2010 to 2024.

Daily returns are calculated from price changes. Portfolio returns are calculated using the baseline ETF weights.

Performance Overview

Portfolio performance is measured using annualized return, annualized volatility, sharpe ratio, and maximum drawdown.

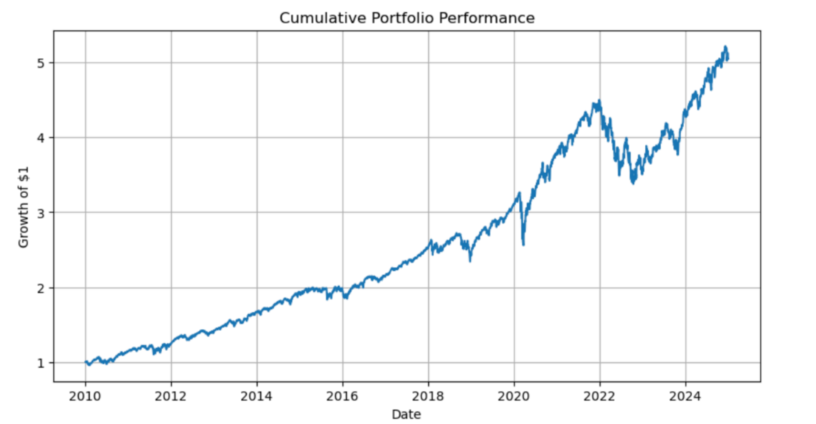
The baseline portfolio generated an annualized return of 11.42% with 11.89% annualized volatility. The sharpe ratio was 0.96.

Metric	Result
Annualized Return	11.42%
Annualized Volatility	11.89%
Sharpe Ratio	0.96
Maximum Drawdown	-24.98%

Cumulative Performance

Cumulative performance shows how one dollar invested in the portfolio grew over time.

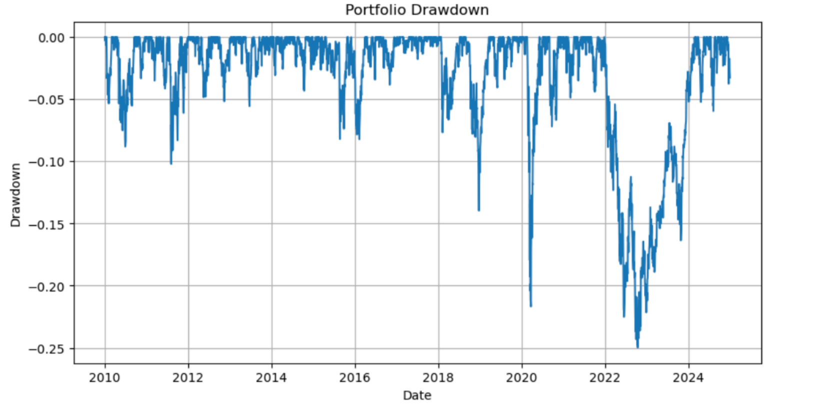
The portfolio grew steadily across the full period, with major declines during market stress periods and a clear recovery after 2022.



Risk and Drawdowns

Drawdown measures the portfolio's decline from a previous high.

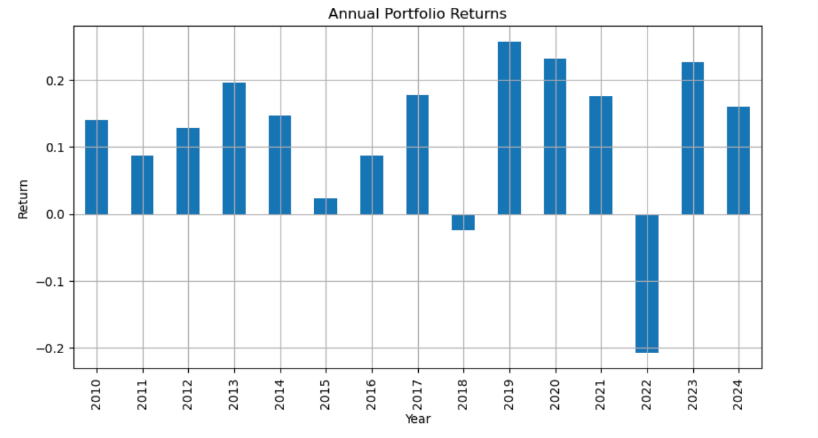
The maximum drawdown was -24.98%, with the largest decline occurring during the 2022 rate hike period.



Annual Returns

Annual returns show how performance changed across calendar years.

The portfolio had mostly positive years, while 2018 and 2022 were negative. The weakest year was 2022, when both stocks and bonds declined.



Diversification Analysis

Correlations are calculated from daily ETF returns.

Equity ETFs were highly correlated with each other. Treasury ETFs had negative correlation with equities, supporting diversification.

Relationship	Correlation
SPY and QQQ	0.93
SPY and IWM	0.88
IEF and TLT	0.92
SPY and TLT	-0.32

Risk Contribution

Risk contribution shows which holdings drove total portfolio volatility.

Most portfolio risk came from equity exposure, especially SPY and QQQ. Treasury ETFs had small negative risk contributions because they moved differently from equities.

ETF	Weight	Risk Contribution
SPY	50.00%	69.30%
QQQ	15.00%	24.28%
IWM	5.00%	8.11%
IEF	20.00%	-0.52%
TLT	10.00%	-1.17%

Benchmark Comparison

The portfolio is compared with SPY to evaluate risk adjusted performance against broad U.S. equities.

SPY had a higher return, but the diversified portfolio had lower volatility and a higher sharpe ratio.

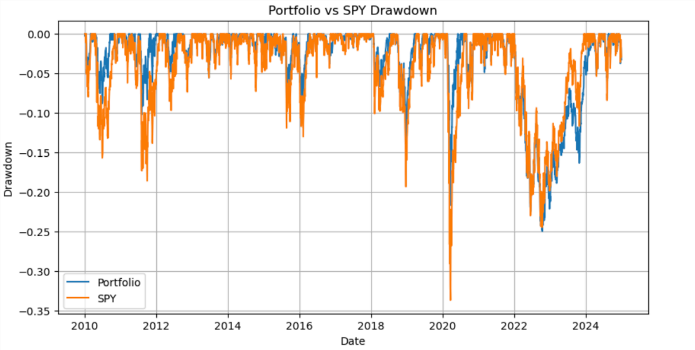
Metric	Portfolio	SPY
Annualized Return	11.53%	14.31%
Annualized Volatility	11.89%	17.05%
Sharpe Ratio	0.97	0.84

Benchmark Drawdowns

Drawdowns are compared to test whether diversification reduced downside risk.

The portfolio had a smaller maximum drawdown than SPY, with a loss of -24.98% compared with -33.72% for SPY.

Metric	Portfolio	SPY
Maximum Drawdown	-24.98%	-33.72%



Historical Stress Testing

Stress testing evaluates portfolio performance during major market declines.

The portfolio lost less than SPY in most stress periods. The main exception was 2022, when both stocks and bonds declined.

Stress Period	Portfolio Return	SPY Return
2011 Debt Ceiling and Eurozone Stress	-6.12%	-16.27%
2015 to 2016 Slowdown	-3.46%	-7.02%
2018 Q4 Selloff	-9.24%	-13.53%
2020 COVID Crash	-21.36%	-33.40%
2022 Rate Hike Shock	-24.53%	-23.83%

Stress Period Drawdowns

Maximum drawdown is also calculated within each stress period.

The diversified portfolio had smaller drawdowns in earlier stress events, but 2022 was different because Treasury exposure also declined.

Stress Period	Portfolio Drawdown	SPY Drawdown
2011 Debt Ceiling and Eurozone Stress	-10.20%	-18.37%
2015 to 2016 Slowdown	-8.22%	-12.82%
2018 Q4 Selloff	-13.51%	-19.20%
2020 COVID Crash	-21.68%	-33.72%
2022 Rate Hike Shock	-24.57%	-24.50%

Regression Exposure

Regression is used to estimate which assets explain portfolio returns.

The portfolio was mainly driven by SPY exposure. IEF and TLT also contributed, but equity risk remained the main driver.

Variable	Coefficient	P Value
SPY	0.7278	0.0000
IEF	0.1940	0.0000
TLT	0.1132	0.0000

Sensitivity Analysis

Sensitivity analysis compares the baseline 70/30 portfolio with 60/40 and 80/20 allocations.

Higher stock exposure increased return and volatility. The 70/30 portfolio stayed between the lower risk 60/40 portfolio and the higher growth 80/20 portfolio.

Allocation	Return	Volatility	Sharpe Ratio	Max Drawdown
60/40	9.73%	10.36%	0.94	-22.79%
70/30	11.42%	11.89%	0.96	-24.98%
80/20	13.05%	13.72%	0.95	-28.76%

Key Findings

The 70/30 portfolio delivered solid risk adjusted performance from 2010 to 2024.

Diversification helped reduce volatility and drawdowns compared with SPY, although it also lowered total return.

Treasury exposure supported the portfolio in most stress periods, but protection weakened in 2022 when stocks and bonds declined together.

The sensitivity analysis confirms the main tradeoff: higher stock exposure increased return potential, while higher bond exposure reduced downside risk.

Overall, the project shows how portfolio risk analysis can support allocation decisions by connecting return, volatility, drawdowns, correlations, and stress testing in one framework.